Pallavi Shakya

Curriculum Vitae

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PallaviShakya

Summary

Computational biologist and nematologist with end-to-end expertise across wet-lab and dry-lab genomics. Experienced in chromosome-scale de novo assembly using PacBio HiFi, Oxford Nanopore, Illumina, and Hi-C, paired with hands-on functional validation (RNAi, RT-qPCR, infection assays). Recent first-author work in *PLoS Pathogens* delivered the most contiguous plant-parasitic nematode assembly to date. Research spans plant pathology, nematology, agricultural biotechnology, and the development of bioinformatics tools and infrastructure.

Education

2021–present **PhD**, *Plant Pathology (Nematology)*, University of California, Davis, Davis, USA

2018–2020 **Master of Science (MSc)**, *Plant Biotechnology*, Wageningen University, Wageningen, Netherlands

2012–2016 **Bachelor of Science (BSc)**, *Biotechnology*, SANN International College – Purbanchal University, Kathmandu, Nepal

Research Experience

2021–present **PhD Researcher**, *UC Davis*, Davis, USA

- **Genome Architecture:** Delivered a chromosome-scale assembly of *Meloidogyne hapla* using PacBio HiFi and Hi-C, resolving complex repetitive regions and non-canonical telomeres.
- **Functional Validation:** Validated candidate effectors and genes of interest using RNAi-mediated gene silencing and RT-qPCR; managed large-scale nematode infection assays to characterize virulence phenotypes.
- **Molecular Techniques:** Executed high-molecular-weight (HMW) DNA and RNA extractions for long-read sequencing (PacBio/ONT) and designed specialized primers for complex genomic regions.
- **Pipeline Engineering:** Developed and deployed reproducible Snakemake workflows for SV discovery and RNA-seq analysis, integrated with Docker for cross-platform portability.
- **Infrastructure Leadership:** Established and managed a lab-wide Apollo genome browser instance, enabling collaborative real-time manual curation for 10+ researchers.
- **Bioinformatics:** Conducted comparative genomics to identify recombination hotspots linked to effector protein evolution in parasitic nematodes.

04/2024–06/2024 **Disease Diagnosis Intern**, *Eskalen Lab, UC Davis*, Davis, USA

- Assisted in culturing, isolating, and molecular identification of pathogens.
- Contributed to the disease diagnosis workflow for specialty crops.

03/2020–09/2020 **Nematology Intern**, *Laboratory of Nematology, Wageningen University*, Wageningen, Netherlands

- Analyzed transcriptome datasets of PCN (*G. pallida*, *G. rostochiensis*).
- Performed differential expression analysis and functional annotation.

Skills

Omics De novo assembly (PacBio, ONT, Hi-C), Annotation, Comparative Genomics, Transcriptomics (RNA-seq, IsoSeq), Phylogenetics

Data Science	Statistical analysis (PCA, clustering, linear/logistic regression), QTL mapping with phenotypic trait validation, exploratory analysis of high-dimensional genomic data; foundational ML coursework (Stanford ML certificate)
Bioinformatics	Snakemake, Docker, Conda, SLURM, Git/GitHub, HPC environments
Programming	R, Python, TensorFlow, Bash, Markdown, LaTeX
Molecular Biology	RNAi, Pathogen assays, RT-PCR, DNA and RNA extraction
Languages	English, Nepali, Hindi, Nepal Bhasa; Intermediate proficiency in Dutch

Publications

2026 Blundell, A. C., Shigekane-Kraft, E., Janakowski, S., Sobczak, M., Dai, D., **Shakya, P.**, et al. Resistance breaking in root-knot nematodes carries a fitness cost associated with defective feeding site development. *bioRxiv* 2026. doi:10.1101/2026.02.17.704924

2025 **Shakya, P.**, Maulana, M. I., Danchin, E. G., Voogt, M. L., Ruitenbeek, S. J. S. van de, Gimeno, J., Taranto, A. P., Blundell, A. C., Despot-Slade, E., Meštrović, N., Mota, A. Z., Dai, D., Williamson, V. M., Sterken, M. G., & Siddique, S. High-resolution genome assembly and linkage mapping in *Meloidogyne hapla* reveal non-canonical telomere repeats and recombination hotspots associated with effector proteins. *PLoS Pathog* 21(11). doi:10.1371/journal.ppat.1013706

2019 Thapa, C., **Shakya, P.**, Shrestha, R., Pal, S., & Manandhar, P. Isolation of polyhydroxybutyrate (PHB) producing bacteria, optimization of culture conditions for PHB production, extraction and characterization of PHB. *Nepal Journal of Biotechnology* 6(1), 62–68. doi:10.3126/njb.v6i1.22339

Honors & Awards

2026 **Cobb Student Travel Award** – Society of Nematologists Conference, USA (**$1,000**)

2026 **Travel Bursary** – European Society of Nematologists (**€900**)

2024 **Mary Allen Travel Award** – University of California, Davis, USA (**$1,500**)

2023 **Bayer CropScience Student Travel Award** – Society of Nematologists Conference, USA (**$600**)

2023 **Jastro-Shields Research Award** – University of California, Davis, USA (**$2,500**)

2022 **Jastro-Shields Research Award** – University of California, Davis, USA (**$3,000**)

2018 **University Fund Wageningen (UFW) Fellowship** – Wageningen University, Netherlands (**€38,600**)

2018 **Anne van den Ban Fund Scholarship** – Wageningen University, Netherlands (**€21,360**)

2018 **HM Queen's Scholarship** – Asian Institute of Technology, Thailand (**1,150,400 Baht; declined**)

2017 **Second Best Poster Presentation Award** – Biotechnology Society Nepal & National Academy of Science and Technology, International DNA Day 2017 (**Rs. 5,000**)

Oral & Poster Presentations

07/2026 ***Invited Speaker***, *High-resolution genome assembly and linkage mapping in Meloidogyne hapla reveal non-canonical telomere repeats and recombination hotspots associated with effector proteins*, Society of Nematology Conference, Baltimore, Maryland, USA

06/2026 ***Plenary***, *High-resolution genome assembly and linkage mapping in Meloidogyne hapla reveal non-canonical telomere repeats and recombination hotspots associated with effector proteins*, European Society of Nematology Conference, Egmond aan Zee, Netherlands

05/2024 ***Poster***, *A chromosome-scale genome assembly reveals novel chromosomal end repeats in a parasitic nematode*, The Biology of Genomes Conference, Cold Spring Harbor Laboratory, New York, USA

08/2023 ***Oral***, *Unraveling the virulence mechanism of Meloidogyne hapla through genomics and functional genetics*, Society of Nematology Conference, Ohio State University, Ohio, USA

05/2022 ***Oral***, *Meloidogyne hapla with improved assembly and genome annotation*, International Congress of Nematology, Antibes, France

04/2016 ***Poster***, *Isolation of Polyhydroxybutyrate (PHB) Producing Bacteria, Optimization of Culture Conditions for PHB Production, Extraction and Characterization of PHB*, National Academy of Science and Technology (National DNA Day), Kathmandu, Nepal

Certifications

2021 **Machine Learning**, *Stanford University (Online)*, USA

2017 **Cambridge English C1 Proficiency**, *British Council*, Kathmandu, Nepal

2017 **German A1 Proficiency**, *Goethe Zentrum*, Kathmandu, Nepal

2017 **Advanced Hands-On Molecular Training**, *Intrepid Nepal & Harvard Medical School*, Kathmandu, Nepal, With focus on DNA extraction, real-time PCR, introductory bioinformatics (data analysis and primer design), and gel electrophoresis

Leadership, Outreach & Mentorship

2026 **Speaker & Event Lead, UC Davis Biodiversity Museum Day.** Organized and hosted a public event on nematode biodiversity and plant health, delivering an accessible talk and facilitating Q&A with K–12 students, growers, and community members.

2026 **Organizing Committee Member, UC Davis Biodiversity Museum Day.** Coordinated public-facing exhibits on plant-parasitic and free-living nematodes, using live demonstrations and hands-on activities to communicate their roles in agriculture and ecosystems.

2022–2024 **Volunteer & Nematode Representative, UC Davis Biodiversity Museum Day & Picnic Day.** Designed and delivered interactive displays on nematode life cycles, host–parasite interactions, and crop damage, highlighting the importance of nematology for sustainable agriculture.

2022–2026 **Mentorship & Training.** Mentored two undergraduate researchers in plant–nematode interaction assays and nematode genomics, guiding experimental design and data interpretation. Trained two PhD students and postdoctoral researchers in RNA-seq and genome assembly workflows, building bioinformatics capacity in nematology-focused research groups.

2022 **Teaching Assistant, Nematology (NEM100), UC Davis.** Supported laboratory sessions, discussions, and grading, helping students connect nematode morphology, biology, and management principles to real-world plant diseases.

2023 **Teaching Assistant, Animal Biology (ABI50), UC Davis.** Assisted with laboratory sessions on nematode parasites of fishes and other taxa, reinforcing core concepts in host–parasite biology and animal health.

2017–2018 **Team Leader, Raleigh International, Nepal.** Spearheaded one year of cross-cultural agricultural development in rural communities. Mentored 14 volunteers (from the UK & Nepal) in community engagement and field safety. Facilitated livelihoods and sustainability training with smallholder farmers and local agro-industries, emphasizing soil health, crop diversification, and sustainable pest management. Strengthened local capacity through farmer-led workshops, linking field realities to long-term agricultural resilience.